Mercy Regional Medical Center

Stroke Service / Department of Neurology
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HOSPITAL DISCHARGE SUMMARY

Patient Name:	Hickle, Brady998
Medical Record Number:	MRN-8821034
Date of Birth:	November 27, 1914
Age:	80 years
Sex:	Male
Admission Date:	July 14, 1995
Discharge Date:	July 20, 1995
Length of Stay:	6 days
Attending Physician:	Margaret T. Sorensen, MD, Neurology
Consulting Services:	Cardiology, Physical Therapy, Occupational Therapy

Admission Diagnosis

Acute ischemic stroke — small-vessel lacunar infarction, right middle cerebral artery (MCA) distribution, presenting with left arm monoparesis.

Discharge Diagnoses

1. Acute ischemic stroke, right MCA territory, lacunar subtype, with residual left upper extremity weakness
2. Silent cerebral microhemorrhages
3. Coronary artery disease
4. Hypertension
5. Type 2 diabetes mellitus
6. Obesity (BMI 30+)
7. Metabolic syndrome
8. Hypertriglyceridemia
9. Chronic kidney disease, stage 1
10. Diabetic renal disease

11. Carcinoma in situ of prostate
12. History of alcohol use disorder

Hospital Course

Presenting History and Examination

Mr. Hickle is an 80-year-old right-handed gentleman with a past medical history notable for coronary artery disease, hypertension, type 2 diabetes mellitus, obesity, chronic kidney disease, and prostate carcinoma in situ, who presented to the Emergency Department on July 14, 1995, at approximately 09:45 hours with sudden onset of left upper extremity weakness. The patient was noted to have difficulty lifting his left arm and reported subjective numbness in the affected limb. Symptom onset was witnessed by the patient's spouse at approximately 08:30 hours while the patient was preparing breakfast. There was no loss of consciousness, headache, visual changes, speech disturbance, or lower extremity involvement at the time of presentation.

On initial neurological examination, the patient was alert and oriented to person, place, and time. Cranial nerves II through XII were grossly intact. Motor examination revealed left upper extremity weakness graded at 3/5 in the deltoid, biceps, triceps, wrist extensors, and intrinsic hand muscles. Lower extremities and right upper extremity strength were 5/5 throughout. Sensory examination demonstrated decreased light touch and pinprick sensation over the left upper extremity. Coordination testing revealed mild dysmetria on left finger-to-nose maneuver. Gait was not assessed initially. National Institutes of Health Stroke Scale (NIHSS) score was calculated at 4.

Acute Stroke Management

Given the presentation within a 2-hour window from symptom onset, the patient was evaluated emergently by the Stroke Service. Non-contrast computed tomography (CT) of the head was obtained at 10:05 hours and demonstrated no evidence of acute intracranial hemorrhage, mass effect, or large territorial infarction. CT angiography of the head and neck demonstrated patent major intracranial vessels without evidence of large vessel occlusion. Serum glucose at presentation was 168 mg/dL.

The patient met inclusion criteria for intravenous tissue plasminogen activator (tPA) therapy and had no absolute contraindications. After a thorough discussion of risks and benefits with the patient and his spouse, informed consent was obtained. Intravenous alteplase (Activase®) was administered at a dose of 0.9 mg/kg (total dose: 63 mg, with 6.3 mg bolus and 56.7 mg infusion over 60 minutes), with infusion commencing at 10:42 hours, approximately 2 hours and 12 minutes from symptom onset.

The patient was monitored closely in the Neuroscience Intensive Care Unit. Blood pressure was maintained with a goal of systolic pressure less than 180 mmHg and diastolic pressure less than 105 mmHg during and after tPA infusion. Neurological examinations were performed every 15 minutes during infusion and hourly thereafter. The patient tolerated the infusion without immediate complication. By 6 hours post-infusion, mild improvement in left upper extremity strength was noted, with power improving to 4-/5 in proximal muscle groups.

Diagnostic Workup

Magnetic resonance imaging (MRI) of the brain with diffusion-weighted imaging (DWI) was obtained on hospital day 2 and revealed a small acute infarct in the right corona radiata consistent with lacunar stroke in the distribution of lenticulostriate perforators. The lesion measured approximately 1.2 cm in maximal diameter. Gradient-echo (GRE) sequences demonstrated multiple foci of susceptibility artifact in bilateral cerebral hemispheres, consistent with chronic microhemorrhages. These were noted predominantly in the basal ganglia, thalami, and subcortical white matter. MR angiography of the intracranial circulation was unremarkable without evidence of stenosis or aneurysm.

Transthoracic echocardiography was performed on hospital day 3 to evaluate for cardioembolic source. The study demonstrated mild left ventricular hypertrophy with preserved ejection fraction estimated at 55%. No intracardiac thrombus, patent foramen ovale, or valvular vegetations were identified. Mild mitral regurgitation and trace tricuspid regurgitation were noted.

Carotid duplex ultrasonography was obtained on hospital day 2 and revealed mild atherosclerotic disease with less than 30% stenosis bilaterally at the carotid bifurcations. No hemodynamically significant stenosis was identified.

Continuous telemetry monitoring was maintained throughout the hospitalization without evidence of atrial fibrillation or other clinically significant arrhythmia. A 12-lead electrocardiogram demonstrated normal sinus rhythm with left ventricular hypertrophy by voltage criteria and nonspecific ST-T wave abnormalities.

Laboratory Findings

Test	Result	Reference Range
Hemoglobin A1c	8.2%	4.0–5.6%
Fasting glucose	154 mg/dL	70–100 mg/dL
Total cholesterol	228 mg/dL	<200 mg/dL
LDL cholesterol	148 mg/dL	<100 mg/dL
HDL cholesterol	38 mg/dL	>40 mg/dL
Triglycerides	356 mg/dL	<150 mg/dL
Creatinine	1.3 mg/dL	0.7–1.3 mg/dL
eGFR	58 mL/min/1.73m²	>60 mL/min/1.73m²
Hemoglobin	11.4 g/dL	13.5–17.5 g/dL
INR	1.0	0.9–1.1

Treatment Course and Clinical Progress

Following thrombolytic therapy, the patient demonstrated gradual neurological improvement. By hospital day 3, left upper extremity strength had improved to 4/5 in all muscle groups. Sensory deficits persisted but were less pronounced. The patient was transitioned from the intensive care unit to the Stroke Unit for continued monitoring and rehabilitation.

Physical therapy and occupational therapy evaluations were obtained. The patient participated in daily therapy sessions with focus on upper extremity strengthening, fine motor coordination, and activities of daily living. The patient demonstrated good effort and motivation. By discharge, he was able to perform most self-care activities with minimal assistance.

Secondary stroke prevention strategies were implemented. Antiplatelet therapy was initiated with aspirin 81 mg daily on hospital day 2 following confirmation of ischemic etiology and resolution of immediate hemorrhagic risk. High-intensity statin therapy was started with atorvastatin 80 mg daily for aggressive lipid management given elevated LDL and multiple vascular risk factors.

Blood pressure management was optimized with continuation of lisinopril 10 mg daily and amlodipine, with escalation of amlodipine from 2.5 mg to 5 mg daily. Blood pressures stabilized with systolic readings in the range of 125–140 mmHg.

Diabetes management was reviewed with the endocrinology service. The patient's home regimen of metformin and insulin was continued with adjustment based on capillary blood glucose monitoring. Hemoglobin A1c of 8.2% indicated suboptimal glycemic control, and the importance of tighter glucose management was emphasized for stroke prevention.

Lifestyle modification counseling was provided, including recommendations for smoking cessation (patient denied current tobacco use), alcohol moderation (given history of alcohol use disorder), Mediterranean diet, weight reduction with goal BMI less than 30, and regular aerobic exercise as tolerated.

Functional Status at Discharge

At the time of discharge, the patient demonstrated residual left upper extremity weakness with strength graded at 4/5 in proximal and distal muscle groups. Fine motor control remained slightly impaired. Gait was mildly unsteady, described as an abnormal gait with decreased left arm swing but without significant lower extremity weakness or ataxia. The patient was able to ambulate independently with close supervision for short distances on level surfaces. Modified Rankin Scale (mRS) at discharge was 2, indicating slight disability with the patient able to look after own affairs without assistance but unable to carry out all previous activities.

Discharge Medications

1. Aspirin 81 mg orally once daily — antiplatelet therapy for stroke prevention
2. Atorvastatin 80 mg orally once daily at bedtime — lipid management
3. Lisinopril 10 mg orally once daily — blood pressure control, renal protection

4. Amlodipine 5 mg orally once daily — blood pressure control
5. Hydrochlorothiazide 25 mg orally once daily — blood pressure control
6. Metformin hydrochloride 500 mg extended-release orally twice daily — diabetes management
7. Insulin human, isophane 70 units/mL / Regular insulin, human 30 units/mL injectable suspension (Humulin 70/30) — 20 units subcutaneously before breakfast and 10 units before dinner
8. Nitroglycerin 0.4 mg/actuation sublingual spray as needed for chest pain — use up to 3 doses 5 minutes apart, call 911 if chest pain persists
9. Ferrous sulfate 325 mg orally once daily — anemia management
10. Epoetin alfa (Epogen) 4,000 units/mL, 1 mL subcutaneously three times per week — anemia management related to chronic kidney disease
11. Naproxen sodium 220 mg orally twice daily as needed for osteoarthritis pain — use with caution, not to exceed 2 weeks continuous use

Discharge Instructions and Patient Education

The patient and family received extensive education regarding stroke pathophysiology, warning signs of recurrent stroke (using the FAST mnemonic: Face drooping, Arm weakness, Speech difficulty, Time to call 911), and the importance of medication adherence. The critical role of blood pressure control, glucose management, and lipid reduction in preventing future cerebrovascular events was emphasized.

The patient was counseled on activity restrictions and gradual resumption of normal activities. He was advised to avoid heavy lifting (greater than 10 pounds) for 2 weeks and to progressively increase physical activity as tolerated. Fall precautions were reviewed given residual weakness and gait abnormality. The patient was instructed to use assistive devices as recommended by physical therapy and to maintain a safe home environment free of tripping hazards.

Dietary counseling was provided with emphasis on a heart-healthy diet: low sodium (less than 2 grams per day), low saturated fat, high in fruits and vegetables, whole grains, and lean proteins. The patient was referred to outpatient nutrition services for further dietary education and meal planning.

The patient verbalized understanding of all discharge instructions and demonstrated ability to correctly identify medications and their purposes.

Follow-Up Arrangements

- **Neurology Clinic (Dr. Sorensen):** Appointment scheduled for 2 weeks post-discharge (August 3, 1995, at 2:00 PM) for stroke follow-up, medication review, and neurological examination. Patient to bring list of all medications and blood pressure log.
- **Primary Care Physician (Dr. Raymond Kessler):** Appointment scheduled for 1 week post-discharge (July 27, 1995) for general medical follow-up, blood pressure check, and review of laboratory studies.

- **Cardiology (Dr. Patricia Nguyen):** Appointment scheduled for 4 weeks post-discharge for cardiac risk factor management and consideration of stress testing.
- **Endocrinology:** Appointment within 6 weeks for diabetes management optimization.
- **Outpatient Physical Therapy:** Referral placed; patient to call Harbor Rehabilitation Services at (617) 555-3200 to schedule within 1 week of discharge for continued upper extremity strengthening and gait training.
- **Occupational Therapy:** Referral placed for outpatient services focused on fine motor skills and activities of daily living.

Laboratory and Imaging Follow-Up

Repeat fasting lipid panel and hepatic function tests recommended in 6 weeks to assess response to high-dose statin therapy. Hemoglobin A1c to be rechecked in 3 months. Repeat carotid duplex ultrasonography in 1 year or sooner if clinically indicated. The patient was provided with a home blood pressure monitor and instructed to record readings twice daily (morning and evening) and bring the log to all follow-up appointments.

Prognosis

The prognosis for functional recovery is cautiously optimistic. The patient has demonstrated improvement from baseline with appropriate acute intervention and has residual deficits that are amenable to ongoing rehabilitation. The presence of silent microhemorrhages on MRI indicates underlying cerebral small vessel disease and places the patient at increased risk for future cerebrovascular events. Aggressive management of vascular risk factors — particularly hypertension, diabetes, and hyperlipidemia — is essential for secondary stroke prevention. Continued participation in physical and occupational therapy will be important for maximizing functional recovery.

Additional Recommendations

- Strict blood pressure control with target less than 140/90 mmHg (consider goal less than 130/80 mmHg if tolerated)
- Intensive glycemic control with target HbA1c less than 7%
- LDL cholesterol goal less than 70 mg/dL
- Weight reduction with goal BMI less than 30
- Regular aerobic exercise at least 30 minutes daily, 5 days per week, as tolerated
- Alcohol moderation; strongly encourage engagement with addiction services given history of alcohol use disorder
- Annual influenza vaccination and pneumococcal vaccination per guidelines
- Consider home safety evaluation by visiting nurse or occupational therapist

Condition at Discharge

Stable, improved from admission. The patient is ambulatory with close supervision, performing most activities of daily living with minimal assistance, and demonstrates good understanding of discharge plan and medication regimen.

Discharge Disposition

Home with family support and outpatient rehabilitation services.

This discharge summary was prepared and attested by:

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